

Target: 6 000 mA h, 12 months, wake every 5 min, BLE-advertise to one gateway per site. No WiFi/LTE on the device.

VERDICT — the spec as written fails, and no measurement can rescue it. Grant every advantage: no battery derate (6 000 mA h → 685 μA mean budget), a free sleep ($I_{\text{sleep}} = 0$), radio off. The duty cycle is fixed by the requirement at 30 s/300 s = 10 %, so $I_{\text{awake,max}} = 685 \mu\text{A}/0.10 = \mathbf{6.85 \text{ mA}}$. **No ESP32 executes code at 6.85 mA** — Espressif's floor (modem sleep, 80 MHz, radio off) is 20 mA to 31 mA. **Over budget by $\geq 3\times$** before the sensor is read. *Three numbers, none of them ours: your mA h, your duty cycle, Espressif's datasheet.*
The fix is time, not milliamps. “Awake 30 s” $\neq$ “CPU runs 30 s” — a $12\times$ difference in battery life, and the spec is silent on which one gets built. **Ceiling: the CPU may run for ≤ 3.9 s per 5 min cycle** [$t_{\text{on,max}} = T(I_{\text{budget}} - I_{\text{sleep}})/(I_{\text{awake}} - I_{\text{sleep}})$].

	Board	CPU-active	Sleep floor	$I_{\text{mean}} \uparrow$	Life (on 5 100 mA h usable)
D0	Devkit	31.2 s	6.5 mA	10.8 mA	20 days — fails on the <i>sleep floor alone</i>
D1	Custom	31.2 s	20 μA	5.01 mA	6.1 weeks — the literal spec, built the obvious way
D2	Custom	2.5 s	20 μA	380 μA	18.4 months — recommended . Short wake; no extra parts
D3	Custom	1.2 s	20 μA	394 μA	17.7 months — literal 30 s window, CPU light-slept; +32.768 kHz xtal
Budget — mean current must not exceed					582 μA (5 100 mA h/8 760 h; 48.5 μA h per cycle)

D2 and D3 land within 5 % of each other — **do not choose between them on battery life**. Take D3 only if one gateway must round-robin many nodes, or if the 30 s window is contractual. **Once the CPU sleeps, boot + init — not the radio — is the largest line item. The two paths have different BOMs: decide before layout.**

1 — Energy per nff operation on ESP32 $\uparrow$ (at the 3.3 V rail)

Operation	t	I	E
nff_init() NVS + hook	12 ms	40 mA	1.6 mJ
WiFi associate + DHCP *	3.1 s	64.7 mA	662 mJ
SNTP cold boot, hard-coded	1.8 s	95 mA	564 mJ
nff_connect() mTLS	1.6 s	110 mA	581 mJ
Heartbeat pub QoS1, 250 B	45 ms	120 mA	17.8 mJ
MQTT keepalive PIN-GREQ	12 ms	115 mA	4.6 mJ
nff_loop() idle tick	0.4 ms	38 mA	0.05 mJ
Cmd ECDSA-P256 verify CPU	55 ms	62 mA	11.3 mJ
↪ ping/diag	18 ms to 30 ms	118 mA	7 mJ to 12 mJ
↪ reboot	0.9 s	88 mA	261 mJ
Crash report 2.5 kB + NVS	180 ms	105 mA	62 mJ
Reconnect one failed TLS	1.9 s	108 mA	677 mJ
Bootstrap announce + rollover	3.2 s	112 mA	1.18 J
OTA total * 1.2 MB	9.9 s	150 mA	4.92 J
OTA probation soak 5 s	5.0 s	82 mA	1.35 J
OTA rollback	0.9 s	88 mA	261 mJ

OTA is not the problem — * 0.41 mA h, 0.007 % of the battery. **WiFi is**. As the SDK ships (it never calls esp_wifi_set_ps()) an associated ESP32 idles at * 49.7 mA: **4.3 days** on this cell. The default 30 s heartbeat alone costs 1 577 mA h/yr, 26 % of the battery. **BLE-only was not a preference — it is the only transport that closes.**

2 — The sleep floor is a board property, not a chip property $\uparrow$
Custom 20 μA: module 10 μA (deep sleep, RTC timer) + low- I_q LDO 1.6 μA + sensor standby & leakage 8 μA. That is 3.4 % of budget.
Devkit 6.5 mA: AMS1117 **5 000 μA** + power LED **1 400 μA** + CP2102 100 μA + the above — **$11\times$ the entire per-cycle budget, asleep.**
A devkit fails the requirement while doing nothing at all. No firmware recovers this; any battery-life figure measured on one is meaningless.

3 — Optimal nff SDK settings under a power constraint

First: the SDK cannot run on this device today. Verified in source — **no BLE** (nff_port.h exposes only nff_port_mqtt.* and https_get_stream; zero NimBLE/bluetooth hits in any firmware) and **no sleep** (no esp_deep_sleep / esp_pm.* anywhere).

Symbol	Set	Why
heartbeat_interval.s	30 300	→ The only shipped power knob (Kconfig max). One heartbeat per wake.
NFF_MQTT_BUFFER_SIZE	1024	A landmine, not a knob. The OTA cmd is 420 B; PubSubClient's 256 B default drops it <i>silently</i> .
NFF_TIMESTAMP_WINDOW.S	300 3600	→ No SNTP when duty-cycled; the RTC's internal RC drifts $\pm 5\%$. At 300 it rejects valid commands.
NFF_LOG_LINES × LINE_LEN	8 × 96	Lives in RTC_NOINIT — retained through deep sleep, so it costs standby current.
NFF_OTA_MAX_DL_RETRIES	4 → 2	Each retry is a fresh TLS handshake. Die fast; retry next cycle.
NFF_OTA_MIN_HEALTHY_MS	5000	Unchanged. The 5 s soak <i>exceeds</i> the 3.9 s ceiling — budget a long-wake exception.

Two hardcoded behaviours must become configurable. (a) The reconnect ladder (nff_core.c:31) saturates at 60 s and retries *forever*: 57 μA h an attempt = **500 mA h/month** against a dead gateway, with no telemetry to show for it — **the SDK's worst battery failure mode**. (b) SNTP (nff_port_esp32_arduino.c:60) is unconditional and blocks nff_connect() up to 20 s on every cold boot: **$5\times$ the entire CPU-active budget**, every cycle.

To build: nff_port_ble_adv.*, nff_port_sleep_deep/light(), nff_run_once(), RTC_NOINIT state carry, a packed 27-B codec, a gateway MQTT bridge. **The PAL is the clean seam — an addition, not a rewrite.**

4 — Optimal ESP32 / BLE parameters

Setting	D2	D3	Why
BT_NIMBLE_ENABLED		y	60 kB less RAM, shorter init. Init is on the critical path.
BT_CLASSIC_ENABLED		n	BR/EDR unused; costs init + RAM.
WiFi		never init	An init'd-but-idle WiFi stack holds the RF/PLL domain up.
BOOTLOADER_SKIP_VALIDATE_IN_DEEP_SLEEP		y	Saves 100 ms to 200 ms <i>every cycle</i> = 2 % to 3 % of budget, for one flag.
BOOTLOADER_LOG_LEVEL DEFAULT_CPU_FREQ_MHZ		NONE 80	UART chatter is pure cost. The window is <i>waiting-bound</i> , not <i>compute-bound</i> .
RTC_CLK_SRC	INT_RC	EXT_CRY	The BOM fork. D2: no xtal (it costs 200 ms/wake). D3: mandatory.
PM_ENABLE TICKLESS_IDLE BTDM_MODEM_SLEEP	+ n +	y	D3 only. All three, or none — miss one and the CPU silently stays awake: you have built D1.
Adv type		NONCONN_IND	Non-connectable <i>and</i> non-scannable: a scannable adv forces RX on after every packet.
Adv interval	100 ms	500 ms	15 events / 1.5 s; 60 events / 30 s.
TX power		9 dBm	Spend the link margin. The radio is on 2.5 ms/100 ms, so max power costs < 1 %; a <i>missed</i> adv costs 60 % of a cycle. Turning the radio down is exactly backwards.
BLE address	static, fixed		Not a rotating RPA — the gateway keys dedup on it.

The 31-byte advert. BLE 4.2 legacy; **no extended advertising on this part — 31 B, not 255**. MSD framing 4 B ⇒ **27 B payload**: ver 1 + seq 2 + flags 1 + fw_id 2 + batt 1 + sensor 16 + CMAC tag 4 — exactly full. **Identity is free:** the 6-B static address is already in the header. The nff heartbeat (250-B JSON) **does not fit, by $8\times$** — the gateway must *reconstruct* it. **Bluedroid's config_adv_data() silently inserts a 3-B Flags AD** — use raw(), or the sensor field shrinks from 16 B to 13 B.

$\uparrow$ **MOCK.** Every current and duration above is invented — plausible, internally consistent, *not measured*. **Real:** everything about the SDK (read from source), the BLE 4.2 / ESP32 capability limits, and the arithmetic. **The verdict does not depend on the mocks** (see box); what the mocks *do* control is the D2/D3 margin — 18

months, or 11, or 30. *Do not promise a customer 12 months on this table yet.* **One measurement converts this sheet:** it all hinges on I_{awake} , which is tens of mA — *inside* the shunt rig’s range (only the 20 μ A sleep floor is not; that needs a PPK2/INA228, and the rig’s own datasheet says so). Flash a bare BLE advertiser — no nff, no WiFi, no sensor — mount per `datasheet.tex` (**DUT USB not connected**), run `nff power measure --for 30`. **Run WIRECHECK and Phase-0 calibration first** — Phase 0 has never been done, so the rig has no R_{eff} and will report a confident, wrong number (wired to *nothing*, it once read 590 mA). **If the BOM is not frozen, change the module:** an ESP32-C3/C6 is the same price and footprint class, with BLE 5.0 (extended adv $\Rightarrow$ 255-B payload, so the nff heartbeat fits *as-is*) and $\approx$ half the active current. *The design closes on the WROOM-32E anyway* — at $1.5\times$ margin instead of $\approx 3\times$. A recommendation, not a blocker. *Derivation: ble-power-budget.tex*